TO DO LIST APPLICATION REPORT

Introduction

The To Do List Application is a web-based productivity application developed to help users manage daily tasks efficiently. The project provides a clean and responsive interface where users can add, update, and delete tasks easily. It was developed using modern web technologies including HTML, CSS, JavaScript, Node.js, and Express.js. The main objective of the project is to understand full-stack web development concepts while creating a practical and user-friendly application.

Abstract

The To Do List Application is a full-stack web project designed to simplify task management through an interactive and responsive interface. The frontend was built using HTML, CSS, and JavaScript, while the backend was developed using Node.js and Express.js. The project also integrates the Groq API to demonstrate external API usage in modern applications. Security measures such as .env configuration and .gitignore were used to protect sensitive API keys. Git and GitHub were used for version control and project management.

Tools and Technologies Used

Frontend Technologies

- HTML5
- CSS3
- JavaScript

Backend Technologies

- Node.js
- Express.js

API and Packages

- Groq SDK
- dotenv
- cors
- body-parser

Development Tools

- Visual Studio Code
- Git
- GitHub
- PowerShell / Terminal

Design Features

- Responsive Layout
- Modern Card-Based UI
- CSS Animations
- Minimal Theme
- Smooth Transitions
- Soft Shadows and Rounded Components

Steps Involved in Building

1. Planned the project structure and features.
2. Designed the frontend using HTML, CSS, and JavaScript.
3. Developed the backend server using Node.js and Express.js.
4. Integrated the Groq API.
5. Secured API keys using .env and .gitignore.
6. Used Git and GitHub for version control and deployment.
7. Tested and debugged the application for smooth functionality.

Conclusion

The To Do List Application successfully demonstrates the development of a modern full-stack web application using frontend and backend technologies. The project helped in understanding important concepts such as responsive UI design, server-side programming, API integration, Git version control, and secure credential management.

The application provides a clean and efficient platform for task management while maintaining a visually attractive user interface. Integration with the Groq API further enhanced the project by demonstrating the practical use of AI services in modern web development.